09 - Ratio Estimator

Catalina Roman

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OBJECTIVE

For a given species, calculate a ratio estimator as a potential mitigation approach to adjust for the loss of sampling coverage caused by offshore wind energy development.

Outputs: abundance indices adjusted with the ratio for each population and simulation of the projection.

METHODOLOGY

A ratio-based estimator was implemented to approximate abundance indices that would have been obtained under full survey coverage when sampling is restricted by offshore wind development. The approach uses information from baseline years, when the status quo survey and the preclusion scenario share the same spatial sampling design except for the exclusion of wind areas.

Abundance indices were first calculated for the status quo and preclusion survey scenarios using a stratified mean estimator. For each simulated population and survey replicate, the number of unique tows was calculated within each stratum and year. Mean catch per tow was then estimated for each stratum and used to compute a stratified abundance index across the survey domain using area-based weights derived from the survey strata.

The first five years of the simulation were used as a baseline period to estimate the relationship between observations from the full survey domain and observations from the restricted domain. During this period, stratum-level mean catches were calculated separately for the status quo survey and for the subset of observations outside wind areas. These values represent the abundance that would be observed with full coverage and with restricted coverage, respectively.

For each population realization, simulation replicate, year, and stratum, an annual ratio was calculated as the ratio of the status quo stratum mean to the corresponding restricted-domain stratum mean. When the restricted-domain mean was zero or missing, the ratio was set to one to avoid undefined expansion factors. These ratios describe the expected relationship between observations obtained from the restricted survey domain and observations that would be obtained if sampling occurred across the full survey footprint.

The ratio values were then applied to tow-level catches from the preclusion survey scenario. For each tow observation, the corresponding ratio was matched by population, simulation, year, and stratum. The observed catch was multiplied by this ratio to obtain an expanded catch value representing an estimate of what would have been observed under full survey coverage.

These expanded catches were then used to calculate stratified abundance indices for the projection period (years 6–15). The same stratified mean estimator and area weights used for the status quo and preclusion scenarios were applied to the expanded observations. This produced an expanded abundance index that approximates the expected survey index under full spatial coverage despite sampling restrictions caused by offshore wind development.

Finally, relative abundance indices were calculated by standardizing annual indices within each population and simulation replicate by the mean index across years. Variance estimates and coefficients of variation were derived from the stratified variance estimator, and confidence intervals were constructed on the log scale and back-transformed to the original scale.

PACKAGES

```
library(tidyverse)
library(here)
library(sdmTMB)
source(here("R", "sim_stratmean_fn.R"))
library(kableExtra)
```

DATA SET UP

Directories

```
sseep.analysis <- "C:/Users/croman1/Desktop/UMassD/sseep-analysis"
dist.dat <- here("data", "rds", "dists")
survdat <- here("data", "rds", "survdat")
surv.prods <- here("data", "rds", "surv-prods")
plots <- here("outputs", "plots")
```

Parameters

```
species <- "scup"
season  <- "fall"
ages      <- 0:7
years     <- 1:15
nsims   <- 1:100
ids     <- sprintf("%03d", nsims)
```

Data

Area weights for each strata

```
strata_wts <- readRDS(here(sseep.analysis, "data", "rds", "active_strata_wts.rds")) |>
  rename(strat = STRATUM)
```

Total survey area

```r
survey_area <- as.integer(sum(strata_wts$Area_SqNm))
```

ABUDNANCE INDEX

calculate the abundance index and relative abundance index for each of the scenarios

Status Quo Survey

```r
#extract the strata that were used to predict spatial distributions in sdmTMB

strat <- map(dist, ~unique(.$strat))

source(here("R/stratmean_fn.R"))

ihat_sq_all <- calc_stratmean(
  surv_list = survdat_sq,
  strata_wts = strata_wts,
  survey_area = survey_area,
  scenario_name = "Status Quo",
  value_col = "n"
)

# Group by pop (and sim if needed) to compute rel_ihat within each population
ihat_sq_all <- ihat_sq_all %>%
  group_by(pop, sim) %>%
  mutate(
    mean_ihat = mean(stratmu),
    rel_ihat = stratmu / mean_ihat,
    rel_var = stratvar / (mean_ihat^2),
    rel_se = sqrt(rel_var),
    rel_cv = rel_se / rel_ihat,
    rel_log_se = sqrt(log(1 + rel_cv^2)),
    rel_ci_lower = exp(log(rel_ihat) - 1.96 * rel_log_se),
    rel_ci_upper = exp(log(rel_ihat) + 1.96 * rel_log_se)
  ) %>%
  ungroup()

ihat_sq_all |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Status Quo Abundance Index (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Status quo stratum means (baseline)

Table 1: Status Quo Abundance Index (first 20 rows)

sim	year	stratmu	stratvar	cv	scenario	pop	mean_ihat	rel_ihat	rel_var	rel_se	rel_cv	rel_log_se	rel_ci_lower	rel_ci_upper
1	1	169.76	6857.84	0.49	Status Quo	1	119.38	1.42	0.48	0.69	0.49	0.46	0.57	3.52
1	2	91.47	1188.84	0.38	Status Quo	1	119.38	0.77	0.08	0.29	0.38	0.36	0.38	1.57
1	3	53.49	237.62	0.29	Status Quo	1	119.38	0.45	0.02	0.13	0.29	0.28	0.26	0.78
1	4	55.21	106.76	0.19	Status Quo	1	119.38	0.46	0.01	0.09	0.19	0.19	0.32	0.67
1	5	66.36	126.50	0.17	Status Quo	1	119.38	0.56	0.01	0.09	0.17	0.17	0.40	0.77
1	6	83.18	728.59	0.32	Status Quo	1	119.38	0.70	0.05	0.23	0.32	0.32	0.37	1.30
1	7	234.53	4531.11	0.29	Status Quo	1	119.38	1.96	0.32	0.56	0.29	0.28	1.13	3.41
1	8	72.12	163.42	0.18	Status Quo	1	119.38	0.60	0.01	0.11	0.18	0.18	0.43	0.85
1	9	191.15	456.41	0.11	Status Quo	1	119.38	1.60	0.03	0.18	0.11	0.11	1.29	1.99
1	10	101.33	343.25	0.18	Status Quo	1	119.38	0.85	0.02	0.16	0.18	0.18	0.59	1.21
1	11	174.87	1070.93	0.19	Status Quo	1	119.38	1.46	0.08	0.27	0.19	0.19	1.02	2.11
1	12	206.03	2617.09	0.25	Status Quo	1	119.38	1.73	0.18	0.43	0.25	0.24	1.07	2.79
1	13	115.85	447.56	0.18	Status Quo	1	119.38	0.97	0.03	0.18	0.18	0.18	0.68	1.38
1	14	90.53	553.98	0.26	Status Quo	1	119.38	0.76	0.04	0.20	0.26	0.26	0.46	1.25
1	15	84.83	756.77	0.32	Status Quo	1	119.38	0.71	0.05	0.23	0.32	0.32	0.38	1.32
2	1	59.05	354.72	0.32	Status Quo	1	108.94	0.54	0.03	0.17	0.32	0.31	0.29	1.00
2	2	59.46	162.43	0.21	Status Quo	1	108.94	0.55	0.01	0.12	0.21	0.21	0.36	0.83
2	3	113.21	2019.08	0.40	Status Quo	1	108.94	1.04	0.17	0.41	0.40	0.38	0.49	2.20
2	4	82.16	413.09	0.25	Status Quo	1	108.94	0.75	0.03	0.19	0.25	0.24	0.47	1.22
2	5	107.27	1430.85	0.35	Status Quo	1	108.94	0.98	0.12	0.35	0.35	0.34	0.50	1.93

```r
sq_mu_y <- map2_dfr(survdat_sq, seq_along(survdat_sq), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 1:5) |>
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct = n_distinct(set),
      mu_sq = sum(n) / towct,
      .groups = "drop"
    )
})

sq_mu_y |>
   dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Status Quo mean years 1-5 (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 2: Status Quo mean years 1-5 (first 20 rows)

pop	sim	year	strat	towct	mu_sq
1	1	1	1010	9	473.00
1	1	1	1020	8	0.38
1	1	1	1030	3	0.00
1	1	1	1040	3	0.00
1	1	1	1050	5	5211.40
1	1	1	1060	10	177.50
1	1	1	1070	3	0.00
1	1	1	1080	3	0.00
1	1	1	1090	5	653.80
1	1	1	1100	10	5.60
1	1	1	1110	3	0.00
1	1	1	1120	3	0.00
1	1	1	1130	9	0.00
1	1	1	1140	3	0.00
1	1	1	1150	3	0.00
1	1	1	1160	10	0.00
1	1	1	1170	3	0.00
1	1	1	1180	3	0.00
1	1	1	1190	8	0.00
1	1	1	1200	4	0.00

Preclusion Survey

```r
ihat_precl_all <- calc_stratmean(
  surv_list = survdat_precl,
  strata_wts = strata_wts,
  survey_area = survey_area,
  scenario_name = "Preclusion",
  value_col = "n"
)

# Group by pop (and sim if needed) to compute rel_ihat within each population
ihat_precl_all <- ihat_precl_all %>%
  group_by(pop, sim) %>%
  mutate(
    mean_ihat = mean(stratmu),
    rel_ihat = stratmu / mean_ihat,
    rel_var = stratvar / (mean_ihat^2),
    rel_se = sqrt(rel_var),
    rel_cv = rel_se / rel_ihat,
    rel_log_se = sqrt(log(1 + rel_cv^2)),
    rel_ci_lower = exp(log(rel_ihat) - 1.96 * rel_log_se),
    rel_ci_upper = exp(log(rel_ihat) + 1.96 * rel_log_se)
  ) %>%
  ungroup()


ihat_precl_all |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
```

```
              caption = "Preclusion Abundance Index (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 3: Preclusion Abundance Index (first 20 rows)

sim	year	stratmu	stratvar	cv	scenario	pop	mean_ihat	rel_ihat	rel_var	rel_se	rel_cv	rel_log_se	rel_ci_lower	rel_ci_upper
1	1	169.76	6857.84	0.49	Preclusion	1	109.95	1.54	0.57	0.75	0.49	0.46	0.62	3.82
1	2	91.47	1188.84	0.38	Preclusion	1	109.95	0.83	0.10	0.31	0.38	0.36	0.41	1.70
1	3	53.49	237.62	0.29	Preclusion	1	109.95	0.49	0.02	0.14	0.29	0.28	0.28	0.85
1	4	55.21	106.76	0.19	Preclusion	1	109.95	0.50	0.01	0.09	0.19	0.19	0.35	0.72
1	5	66.36	126.50	0.17	Preclusion	1	109.95	0.60	0.01	0.10	0.17	0.17	0.43	0.84
1	6	66.44	878.12	0.45	Preclusion	1	109.95	0.60	0.07	0.27	0.45	0.43	0.26	1.39
1	7	176.01	4335.36	0.37	Preclusion	1	109.95	1.60	0.36	0.60	0.37	0.36	0.79	3.25
1	8	68.47	50.40	0.10	Preclusion	1	109.95	0.62	0.00	0.06	0.10	0.10	0.51	0.76
1	9	172.34	1014.29	0.18	Preclusion	1	109.95	1.57	0.08	0.29	0.18	0.18	1.09	2.24
1	10	96.77	166.09	0.13	Preclusion	1	109.95	0.88	0.01	0.12	0.13	0.13	0.68	1.14
1	11	239.33	1326.31	0.15	Preclusion	1	109.95	2.18	0.11	0.33	0.15	0.15	1.62	2.93
1	12	237.72	6959.44	0.35	Preclusion	1	109.95	2.16	0.58	0.76	0.35	0.34	1.11	4.22
1	13	67.26	588.98	0.36	Preclusion	1	109.95	0.61	0.05	0.22	0.36	0.35	0.31	1.21
1	14	51.60	697.54	0.51	Preclusion	1	109.95	0.47	0.06	0.24	0.51	0.48	0.18	1.21
1	15	37.00	48.39	0.19	Preclusion	1	109.95	0.34	0.00	0.06	0.19	0.19	0.23	0.48
2	1	59.05	354.72	0.32	Preclusion	1	99.61	0.59	0.04	0.19	0.32	0.31	0.32	1.09
2	2	59.46	162.43	0.21	Preclusion	1	99.61	0.60	0.02	0.13	0.21	0.21	0.39	0.90
2	3	113.21	2019.08	0.40	Preclusion	1	99.61	1.14	0.20	0.45	0.40	0.38	0.54	2.41
2	4	82.16	413.09	0.25	Preclusion	1	99.61	0.82	0.04	0.20	0.25	0.24	0.51	1.33
2	5	107.27	1430.85	0.35	Preclusion	1	99.61	1.08	0.14	0.38	0.35	0.34	0.55	2.11

Preclusion stratum means for baseline

```
precl_mu_y <- map2_dfr(survdat_sq, seq_along(survdat_sq), function(surv, pop_num) {
  surv |>
    as_tibble() |>
    filter(strat %in% unlist(strat), year %in% 1:5, AREA_CODE != 1) |>
    group_by(pop = pop_num, sim, year, strat) |>
    summarise(
      towct = n_distinct(set),
      mu_precl = sum(n) / towct,
      .groups = "drop"
    )
})

precl_mu_y |>
   dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Preclusion mean years 1-5 (first 20 rows)") |>
   kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 4: Preclusion mean years 1-5 (first 20 rows)

pop	sim	year	strat	towct	mu_precl
1	1	1	1010	8	527.25
1	1	1	1020	8	0.38
1	1	1	1030	3	0.00
1	1	1	1040	3	0.00
1	1	1	1050	2	1379.50
1	1	1	1060	9	33.56
1	1	1	1070	3	0.00
1	1	1	1080	3	0.00
1	1	1	1090	3	2.33
1	1	1	1100	10	5.60
1	1	1	1110	3	0.00
1	1	1	1120	3	0.00
1	1	1	1130	9	0.00
1	1	1	1140	3	0.00
1	1	1	1150	3	0.00
1	1	1	1160	10	0.00
1	1	1	1170	3	0.00
1	1	1	1180	3	0.00
1	1	1	1190	8	0.00
1	1	1	1200	4	0.00

Ratio calculation (y_tot/y_preclusion)

```
# ratio_tbl <- sq_mu |>
#   left_join(precl_mu, by = c("pop", "sim", "strat"))
#
# ratio_tbl <- ratio_tbl |>
#   mutate(ratio = ifelse(is.na(mu_precl) | mu_precl == 0, 1, mu_tot / mu_precl))
#



ratio_tbl <- sq_mu_y |>
  left_join(precl_mu_y, by = c("pop", "sim", "year", "strat")) |>
  mutate(
    ratio_y = ifelse(is.na(mu_precl) | mu_precl == 0, 1, mu_sq / mu_precl)
  ) |>
  group_by(pop, sim, year ,strat) |>
  summarise(
    ratio = mean(ratio_y, na.rm = TRUE),
    .groups = "drop"
  )

ratio_tbl |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Ratio (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("H"), font_size = 8)
```

Table 5: Ratio (first 20 rows)

pop	sim	year	strat	ratio
1	1	1	1010	0.90
1	1	1	1020	1.00
1	1	1	1030	1.00
1	1	1	1040	1.00
1	1	1	1050	3.78
1	1	1	1060	5.29
1	1	1	1070	1.00
1	1	1	1080	1.00
1	1	1	1090	280.20
1	1	1	1100	1.00
1	1	1	1110	1.00
1	1	1	1120	1.00
1	1	1	1130	1.00
1	1	1	1140	1.00
1	1	1	1150	1.00
1	1	1	1160	1.00
1	1	1	1170	1.00
1	1	1	1180	1.00
1	1	1	1190	1.00
1	1	1	1200	1.00

Calculate expanded index

```
survdat_precl_ratio <- survdat_precl

for (i in seq_along(survdat_precl_ratio)) {
  survdat_precl_ratio[[i]] <- survdat_precl_ratio[[i]] |>
    as_tibble() |>
    mutate(pop = i) |>
    left_join(
      ratio_tbl,
      by = c("pop", "sim", "year","strat")
    ) |>
    mutate(
      ratio = ifelse(is.na(ratio), 1, ratio),
      n_ratio = n * ratio
    )
}

ihat_precl_exp_all <- calc_stratmean(
  surv_list = survdat_precl_ratio,
  strata_wts = strata_wts,
  survey_area = survey_area,
  scenario_name = "Preclusion_expanded",
  value_col = "n_ratio",
  years = 6:15
)

ihat_precl_exp_all <- ihat_precl_exp_all %>%
```

```r
  group_by(pop, sim) %>%
  mutate(
    mean_ihat = mean(stratmu),
    rel_ihat = stratmu / mean_ihat,
    rel_var = stratvar / (mean_ihat^2),
    rel_se = sqrt(rel_var),
    rel_cv = rel_se / rel_ihat,
    rel_log_se = sqrt(log(1 + rel_cv^2)),
    rel_ci_lower = exp(log(rel_ihat) - 1.96 * rel_log_se),
    rel_ci_upper = exp(log(rel_ihat) + 1.96 * rel_log_se)
  ) %>%
  ungroup()



ihat_precl_exp_all |>
  dplyr::slice_head(n = 20) |>
  knitr::kable(format="latex", booktabs=TRUE, digits=2,
               caption = "Preclusion expanded (first 20 rows)") |>
  kableExtra::kable_styling(latex_options = c("hold_position", "scale_down"))
```

Table 6: Preclusion expanded (first 20 rows)

sim	year	stratmu	stratvar	cv	scenario	pop	mean_ihat	rel_ihat	rel_var	rel_se	rel_cv	rel_log_se	rel_ci_lower	rel_ci_upper
1	6	66.44	878.12	0.45	Preclusion_expanded	1	121.29	0.55	0.06	0.24	0.45	0.43	0.24	1.26
1	7	176.01	4335.36	0.37	Preclusion_expanded	1	121.29	1.45	0.29	0.54	0.37	0.36	0.71	2.95
1	8	68.47	50.40	0.10	Preclusion_expanded	1	121.29	0.56	0.00	0.06	0.10	0.10	0.46	0.69
1	9	172.34	1014.29	0.18	Preclusion_expanded	1	121.29	1.42	0.07	0.26	0.18	0.18	0.99	2.03
1	10	96.77	166.09	0.13	Preclusion_expanded	1	121.29	0.80	0.01	0.11	0.13	0.13	0.62	1.03
1	11	239.33	1326.31	0.15	Preclusion_expanded	1	121.29	1.97	0.09	0.30	0.15	0.15	1.47	2.65
1	12	237.72	6959.44	0.35	Preclusion_expanded	1	121.29	1.96	0.47	0.69	0.35	0.34	1.00	3.82
1	13	67.26	588.98	0.36	Preclusion_expanded	1	121.29	0.55	0.04	0.20	0.36	0.35	0.28	1.10
1	14	51.60	697.54	0.51	Preclusion_expanded	1	121.29	0.43	0.05	0.22	0.51	0.48	0.17	1.10
1	15	37.00	48.39	0.19	Preclusion_expanded	1	121.29	0.31	0.00	0.06	0.19	0.19	0.21	0.44
2	6	91.62	2491.32	0.54	Preclusion_expanded	1	107.30	0.85	0.22	0.47	0.54	0.51	0.31	2.32
2	7	190.73	309.79	0.09	Preclusion_expanded	1	107.30	1.78	0.03	0.16	0.09	0.09	1.48	2.13
2	8	57.36	78.75	0.15	Preclusion_expanded	1	107.30	0.53	0.01	0.08	0.15	0.15	0.40	0.72
2	9	134.44	1739.29	0.31	Preclusion_expanded	1	107.30	1.25	0.15	0.39	0.31	0.30	0.69	2.27
2	10	74.33	223.05	0.20	Preclusion_expanded	1	107.30	0.69	0.02	0.14	0.20	0.20	0.47	1.02
2	11	151.58	1489.96	0.25	Preclusion_expanded	1	107.30	1.41	0.13	0.36	0.25	0.25	0.86	2.31
2	12	228.50	10861.93	0.46	Preclusion_expanded	1	107.30	2.13	0.94	0.97	0.46	0.43	0.91	4.99
2	13	23.81	39.69	0.26	Preclusion_expanded	1	107.30	0.22	0.00	0.06	0.26	0.26	0.13	0.37
2	14	38.49	186.70	0.35	Preclusion_expanded	1	107.30	0.36	0.02	0.13	0.35	0.34	0.18	0.70
2	15	82.17	488.87	0.27	Preclusion_expanded	1	107.30	0.77	0.04	0.21	0.27	0.26	0.46	1.29